Sahaj Bhadja

(213) 284-4651 sahaj.bhadja@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Backend and cloud-platform engineer with 2+ years building **distributed services, REST APIs, data pipelines, and containerized infrastructure** across AWS, Azure, and GCP, emphasizing networking, security, observability, and production reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**

Master of Science in Computer Science *GPA: 3.75/4*

Coursework: Database Systems, Advanced NLP, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**

Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*

Coursework: Software Engineering, Data Structures, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**

Software Engineer *Los Angeles, CA*

- Built an Azure-backed multi-tenant platform with **normalized schemas, row-level isolation, and indexed queries**, reducing dashboard latency from 1.2 seconds to 350 milliseconds for 500+ daily campus users.
- Reworked asynchronous FastAPI orchestration with **service-to-service communication, state transitions, retry boundaries, failure isolation, metrics, and logs**, tripling throughput for 1K+ concurrent workloads across stable production release cycles.

Grad8 **June 2025 – August 2025**

Software Engineer Intern *Los Angeles, CA*

- Developed React and Flask workflows backed by **REST APIs and client-server data models**, supporting 150+ instructors and 10K+ daily production requests for uploads, grading history, and feedback.
- Implemented concurrent Python ingestion with **durable processing, validation, error classification, and operational debugging** for PDF, DOCX, ZIP, OCR, and multimodal workloads, reducing peak feedback turnaround 60% daily.

Aasma Technology Solutions **August 2023 – July 2024**

Software Developer Engineer - (SecureSync) *India*

- Designed cross-platform APIs and AWS data pipelines using **microservice boundaries, Glue, S3, and secure integrations**, improving response times 30% while reducing technical debt 20% across critical distributed workflows.
- Automated 20+ unit and integration suites using PyTest, Mockito, Jenkins, and virtual machines, strengthening **service compatibility, CI/CD validation, production debugging, and cross-platform reliability** across distributed customer services.

Aasma Technology Solutions **May 2023 – August 2023**

Full Stack Developer Intern - (VibrantChem) *India*

- Built eight modular services for authentication, resources, billing, and permissions, synchronizing 2M+ records through **REST APIs, WebSockets, relational schemas, authorization controls, and event-driven integrations** across distributed locations.
- Automated Docker and Jenkins releases, reducing deployment time 50% while applying **Sentry monitoring, structured logging, alerting, incident diagnosis, and rollback-ready deployments** across eight customer-facing production service environments.

LST Capitals **November 2022 – January 2023**

Web Developer Intern *India*

- Translated client workflows into technical specifications for 12+ integrations, mapping bookings, payments, coupons, and statuses into **reliable Flask APIs, Firebase schemas, webhooks, and auditable production data flows**.
- Built responsive interfaces and payment services processing 50K+ records with sub-200-millisecond responses, adding **input validation, authentication, failure handling, and integration testing** for dependable customer transactions across platforms.

Skills

Backend: Go, Python, FastAPI, Flask, Java, Node.js, REST APIs, gRPC, AsyncIO, Microservices, Service Registration, Orchestration, API Design

Networking & Security: HTTP, TLS, TCP/IP, DNS, Client-Server Communication, Proxies, Gateways, Secure Tunneling, PrivateLink, Private Service Connect, Certificates, Secrets Management, Encryption

Cloud & Platform: AWS, GCP, Azure, Kubernetes, Docker, Helm, Terraform, ECS, Fargate, VPC, Load Balancing, Cloud Networking, Infrastructure as Code

Data & Observability: Kafka, Redis, Pub/Sub, PostgreSQL, Event-Driven Systems, Datadog, OpenTelemetry, Prometheus, Metrics, Distributed Tracing, Structured Logging, Alerting, SLOs

Engineering: TypeScript, React, Git, Linux, CI/CD, Jenkins, GitHub Actions, Unit Testing, Integration Testing, PyTest, Debugging, Code Reviews, Incident Response, On-Call Operations

Projects

Secure Data Bridge Gateway | *Go, AWS PrivateLink, GCP PSC, TLS, gRPC, Terraform*

- Designed a **secure proxy and relay gateway** connecting cloud workloads to private data sources through AWS PrivateLink and GCP Private Service Connect, with TLS, registration, routing, health checks, and gRPC control APIs.

Cloud Transport Control Plane | *Go, Kubernetes, Helm, Kafka, Redis, OpenTelemetry*

- Built a **distributed transport orchestration service** managing registration, configuration, routing, and connection lifecycle across Kubernetes data planes, with Kafka events, Redis state, retries, alerts, metrics, logs, and distributed traces.

Linguistix | *Python, AWS ECS, Fargate, Docker, CodePipeline, CI/CD*

- Deployed a **containerized Python language service** on AWS ECS and Fargate, automating image builds and releases through CodePipeline while documenting repeatable operations, deployment checks, and production troubleshooting workflows.